

A Data-Driven Pipeline for Extracting Spontaneous and Prototypical Emotional Body Behaviors for Virtual Human Animation

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ABSTRACT

In virtual reality, the perceived authenticity of virtual humans depends on both facial expression and body posture. However, most emotion-animation pipelines still rely on acted motions, which miss the subtle, spontaneous micro-reactions that are critical for nonverbal cues. To bridge this gap, we record lay participants reacting spontaneously to ten emotions induced by video. Addressing the challenge of isolating subtle, non-exaggerated reactions, we propose a multi-criteria consensus filter that combines temporal crowd synchrony and structural similarity. This pipeline drastically reduces the volume of raw motion-capture data, remains tunable through its hyperparameters to produce different operating points, and reveals a first-order taxonomy of emotional responses based on their robustness to filtering. Overall, this work provides a methodological foundation for building a high-fidelity dataset of spontaneous emotional body behaviors, with the long-term goal of driving more expressive and believable virtual humans.

Index Terms: Virtual humans, Spontaneous behavior, Motion capture, Compositional data, Affective computing, Clustering

1 INTRODUCTION

In socially constrained situations, individuals often invest substantial effort in regulating their facial expressions, while their bodies still betray their true feelings. Fundamental work on *nonverbal leakage* has demonstrated that bodily movements are harder to suppress than facial expressions and convey reliable, albeit subtle, affective cues [8]. More recent experimental evidence suggests that, during intense emotional moments, the body can be even more reliable than the face: observers classify a tennis player's outcome (win vs. loss) more accurately from the body posture than from the face, especially when visual information is ambiguous (body and face are not congruent) [3]. Thus, the body remains a particularly authentic channel for emotion, even when the face is controlled. These findings present a clear opportunity for the Virtual Reality community.

However, to create believable virtual humans, animation cannot rely solely on facial expressions or scripted, stereotypical gestures. Currently, most virtual agents are driven by motion capture performed by actors. While technically clean, such data is often limited by the “acting bias”: actors tend to produce stereotypical or exaggerated gestures intended to be clearly readable, often at the expense of the subtle, involuntary cues that are specific to authentic reactions [4]. Spontaneous behavior is preferable but much more

difficult to extract: identifying which movements are truly prototypical (that characterizes an emotion) among the inherent noise and heterogeneity of individual reactions remains a key challenge.

This Work-in-Progress paper is the first part of an ongoing multidisciplinary project, grouping psychology, data science and computer graphics, and aiming to build a foundation for endowing 3D avatars with expressive dynamic postures from real human behavior. Unlike traditional approaches, we captured the spontaneous reactions of lay participants using an optical motion capture system, based on passive markers, in a standing, passive viewing context. The data-driven strategy raises a major technical challenge: how to automatically extract clean, prototypical emotional movements from dozens of hours of raw motion data that are noisy, heterogeneous, and dominated by task-unrelated adjustments or immobility?

To address this, we introduce a “Consensus-based Filtering Pipeline”. Instead of manually cleaning data, we take advantage of the wisdom of the crowd: we assume that a movement is a valid emotional prototypical reaction only if it is performed: (1) synchronously by a significant subset of participants (stimulus causality) and (2) with a similar corporal structure (semantic consistency). To implement this pipeline, we propose a dedicated algorithm combining temporal windowing and density-based clustering in a compositional data space. This process filters massive raw recordings (44.5 hours) to retain only a concise set of promising prototypes for animating emotionally expressive virtual humans.

This paper is organized as follows. Section 2 reviews related work on emotional body movement. Section 3 details the experimental protocol and data acquisition. Section 4 describes the proposed consensus-based filtering method. Section 5 presents preliminary analysis on pipeline sensitivity. Finally, Section 6 concludes the paper and presents future work. Higher-resolution figures are provided in the supplementary material.

2 RELATED WORKS

As virtual humans approach photorealism, users become increasingly sensitive to subtle motion artifacts, a phenomenon often referred to as the “Uncanny Valley”.¹ Buisine et al. [5] showed that adding even neutral idle motion to a virtual agent improves emotion detection and perceived quality compared to a fully still body, and that body-face congruence further enhances perceived realism. Crenn et al. [7] proposed extracting body expressions from acted datasets by synthesizing a “neutral” motion and computing the residual difference. However, this approach relies on a separate action (e.g., walking) to derive a neutral baseline. While effective for such active tasks, it is inapplicable to passive contexts. Beyond animation and focusing on feature validity, Mahfoudi et al. [11] validated 33 theoretical motion descriptors on acted datasets, but explicitly highlighted the critical need to validate them on spontaneous behaviors. To address it, Chen et al. [6] introduced SMG, a large dataset containing 8 hours of spontaneous micro-gestures. However, SMG focuses specifically on stress scenarios and relies on video-based pose estimation, limiting fine-grained 3D analysis.

¹This hypothesis suggests that in highly detailed characters, any lack of natural body motion can severely compromise perceived authenticity.

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To the best of our knowledge, there is currently no 3D dataset or methodology that jointly addresses these constraints: spontaneous standing postures of lay persons in a passive viewing context, fine-grained whole-body motion, and the extraction of emotionally informative motions and micro-gestures from raw data. Our work explicitly targets this gap. We develop a pipeline to isolate prototypical emotional corporal responses. The process aims to support our long-term objectives: (1) emotion analysis and prediction and (2) the generation of credible nonverbal behavior in virtual humans, ultimately enabling the evaluation of whether emotional states can be perceived solely from bodily cues.

3 EXPERIMENTAL SETUP AND DATA

3.1 Experimental Design

We report on data acquired from the first 68 lay participants (balanced by gender, aged 18-40, target $N_{\text{participants}} = 80$) recruited from the general population. The protocol was designed in collaboration with psychology experts to ensure ecological validity while adhering to the constraints of optical motion capture.

Participants stand in the capture area for a session lasting approximately one hour. They are instructed to watch the screen naturally, without performing any explicit task (passive viewing). The study covers ten discrete emotions, sampled across Russell's valence-arousal circumplex model [12] (positive vs. negative, calm vs. excited): Fear, Disgust, Anger, Anxiety, Sadness, Amusement, Happiness, Interest, Relief and Tenderness.

To ensure robust emotional induction, psychology experts selected the three most effective clips out of five per emotion, based on a validation study involving a separate group of 64 participants (using facial expression analysis and subjective self-reports). Therefore, during a session, each participant watches a total of 30 clips (3×10), corresponding to approximately 40 minutes of stimulation. To reinforce the targeted emotional state, clips are grouped into continuous emotion blocks. We apply a two-level random scheme: both the order of the clips within the block and the order of the emotional blocks are randomized. Between blocks, a debriefing phase restores the participant's neutral baseline and verifies stimulus comprehension: participants who fail this check are excluded. Moreover, to avoid any physical fatigue caused by standing still, a 5-10 minute break is scheduled mid-session. This protocol was approved by the institutional ethics committee and all participants provided their consent.

3.2 Motion Capture System

Whole-body motion is recorded using a high-accuracy optical system manufactured by ART (Advanced Realtime Tracking) [1], configured to reconstruct a full 3D skeletal model. The system uses passive retro-reflective targets to track major body joints. The acquired raw data consist of 20-joint 3D positions and 18 bone rotations. These rotation signals constitute the raw input for the pipeline described in the next section.

4 METHOD

Extracting animation-ready micro-gestures captured from the spontaneous reactions of passive, standing participants essentially boils down to a signal-to-noise ratio problem, since most of the recorded data consists of immobility, postural adjustments of balance, or idiosyncratic gestures unrelated to an emotional stimulus.

To filter the raw data, we propose a four-step pipeline: (i) defining a motion representation that is consistent with rigid-articulated-body kinematics and avoids redundant information, (ii) segmenting continuous Quantity-of-Motion (QoM) into discrete movement events at the limb level, (iii) isolating time windows that are plausibly driven by the emotional stimulus rather than by individual noise, and (iv) grouping movements that are not only synchronous but also

structurally similar, so that they can serve as potential prototypical candidates for virtual human animation.

4.1 Rotational Motion Representation

The ART system outputs per-frame global 3D joint positions and local bone rotations. Our first objective is to eliminate "static background", as any analysis run directly on the full time series would be dominated by long periods of quasi-immobility.

In 3D space, the motion of an object can be decomposed into a translation and a rotation. However, we base the QoM on local bone rotations only. This choice is motivated by: (1) **biomechanical consistency** (relative bone motion is purely rotational for an articulated rigid-body), (2) **completeness** (rotations capture axial twists missed by positions), and (3) **morphological robustness** (bone rotations are scale-invariant, allowing direct comparison across participants, without normalization).

Based on this, we define our main signal: the Rotational QoM. We first calculate instantaneous relative displacements for all 18 bones by differentiating local quaternions. To reduce dimensionality and simplify processing, we aggregate these segments into six body parts (Head-Neck, Torso, Left/Right Arms, Left/Right Legs), producing a single 1D mean motion signal per body member.

4.2 Intra-Participant Motion Segmentation

The second stage of our pipeline segments the rotational QoM into discrete movement sequences at the limb level. Concretely, we proceed as follows:

- **Signal partitioning:** For each of the six QoM time-series, we run the PELT[10] (Pruned Exact Linear Time) algorithm to automatically locate change-points. This splits the continuous signal into contiguous blocks, considered statistically stationary, for which we compute a single mean QoM value.
- **Motion Segmentation:** To distinguish true movement from postural noise and immobility sequences without manual tuning, we use Otsu's thresholding. It dynamically calculates an optimal separation threshold (T_{on}) from "a background sway or a static sequence" and "active motion". For each retained movement interval, we compute the Area Under the QoM Curve (AUC) above T_{on} . It quantifies the cumulative motion intensity and serves as the basis for determining the member's relative contribution to the final motion signature.

This process yields, for each segment, cleaned movement intervals that concentrate the data on active phases, effectively removing the static background. Fig. 1 illustrates the raw data of one participant segmented in 6 QoM signals with their detected motion sequences highlighted. Notably, 72.1% of the raw recordings (approximately 32 out of 44.5 hours) are discarded at this stage as no body part exhibits motion, resulting in a preliminary set of 8,031 movements (38.6% of which are shorter than 2 seconds).

All thresholds (colored horizontal lines in Fig. 1) are defined dynamically from each QoM signal's statistics, rather than being fixed globally, so they adapt to the local dynamics of each participant and body part. For subsequent stages, we merge bilateral pairs (left/right arms and legs), resulting in four body parts: Head-Neck, Torso, Arms, and Legs. This makes the analysis invariant to handedness and reduces lateral-dominance bias.

4.3 Inter-Participant Temporal Synchrony Filter (N)

After intra-participant segmentation, motionless intervals are removed, yet the dataset is populated by a large number of idiosyncratic and parasitic motions (e.g., random comfort adjustments) unrelated to the stimulus. To isolate emotionally driven responses, we leverage the crowd size by looking at the entire group simultaneously. Our main assumption is that stimulus-induced reactions are

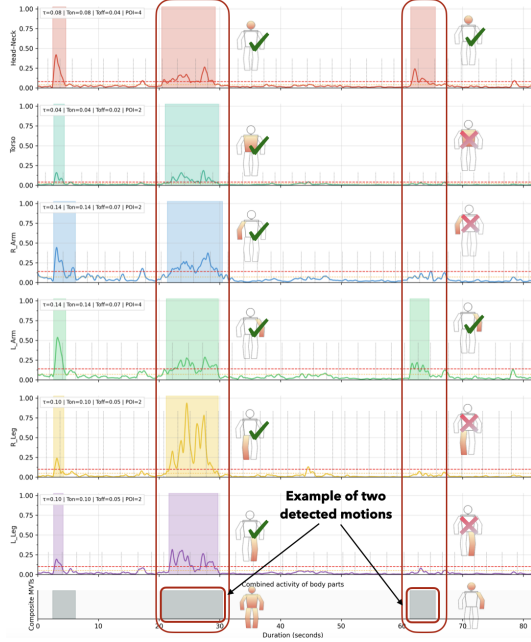


Figure 1: Example of rotational Quantity-of-Motion (QoM) signals for the six body segments of a single participant and stimulus. Horizontal colored lines indicate the dynamic thresholds for each segment. Signal intervals exceeding these thresholds (highlighted regions) are retained as motion sequences.

at least partly synchronous among participants watching the same clip, whereas parasitic movements are more asynchronous.

We therefore introduce a temporal synchrony filter based on a threshold N , representing the minimum number of participants required for synchrony. Using a sliding window w , we count distinct participants exhibiting motion within w . An interval is retained only if this count is greater than or equals to N , otherwise it is discarded.

4.4 Inter-Participant Structural Homogeneity Filter (M)

Temporal synchrony (N) acts as a causality filter, ensuring participants react when the stimulus occurs. However, it does not guarantee that participants perform comparable movements. The reaction set is usually multimodal. For example, in a fear-inducing moment with 25 responsive participants: 15 participants might recoil (Torso+Legs dominance), 3 might shield their faces (Arms dominance), 3 turn the head (Head dominance), while 4 others simply perform unrelated comfort adjustments (noise). Treating such a heterogeneous window as a single group is methodologically flawed: it conflates very distinct behaviors and fails to filter out coincidental noise, leading to false positives. To isolate usable movements, we must move from simple synchrony to structural similarity, ensuring that we compare only movements of the same general type. To achieve this, we apply the following three-stage pipeline to each time window w selected by the N -filter:

- **Kinematic Distribution Signatures:** To characterize the structure of a movement (regardless of its intensity), we compute a 4-dimensional vector S :

$$S = [p_{\text{head}}, p_{\text{torso}}, p_{\text{arms}}, p_{\text{legs}}] \text{ with } p_x = \frac{AUC(QoM_x)}{\sum_{i=1}^4 AUC(QoM_i)}$$

where each component p_x represents the proportion of total rotational displacement contributed by that body part (sum-

ming to 1). For example, a defensive “face shielding” gesture would present a signature dominated by the arms (e.g., [0.3, 0.05, **0.6**, 0.05]). This signature encodes *how* the movement is distributed across the body rather than its magnitude.

- **Metric Adaptation:** Since these components form compositional data (they live on a simplex), standard Euclidean distances are statistically invalid and prone to spurious correlations (the “closure problem”) [9]. To correctly compare the ratios between body parts, we map the signatures into an unconstrained space using the Isometric Log-Ratio (ILR) transformation. This provides an Aitchison geometry [2] in which distances capture proportional changes: for example, a doubling (10% $\rightarrow$ 20%) is treated equivalently to (40% $\rightarrow$ 80%), whereas the linear step (80% $\rightarrow$ 90%) represents a much smaller relative change.
- **Density Clustering:** To enforce structural similarity of motions, we apply DBSCAN (density-based spatial clustering of applications with noise) to these ILR coordinates. This algorithm is well-suited to our problem because it: (i) does not require pre-specifying the number of clusters, (ii) identifies dense cores of similar behaviors so that only motions with comparable signatures form a cluster, (iii) automatically labels rare or unique reactions as noise.

Finally, a cluster is validated at this stage only if it contains motions from at least M participants.

The output of this pipeline is a set of Meta-Movements: distinct behavioral patterns that are (i) temporally synchronized with the emotional stimulus and (ii) structurally shared across the group.

5 EVALUATION AND SENSITIVITY ANALYSIS

To evaluate experimentally the filtering behavior and quantify the impact of crowd synchrony and structural similarity, we conducted an exploratory analysis on the dataset acquired from $N_{\text{part}} = 68$ participants. We executed the full pipeline over a grid of 444 hyperparameter configurations, varying $N \in [4, 42]$ and $M \in [4, 34]$, with the constraint $M \leq N$.

5.1 Global Sensitivity Analysis

To better understand how the retained volume reacts to small changes in the hyperparameters, we compute two “sensitivity” maps. For each pair (N, M) , we measure the local percentage change in the number of movements when increasing one parameter by one unit while keeping the other fixed. The structural sensitivity map Δ_M (symmetrically Δ_N) is defined as:

$$\Delta_M(N, M) = 100 \times \frac{V(N, M+1) - V(N, M)}{V(N, M)}, \quad (1)$$

where $V(N, M)$ denotes the number of movements retained for (N, M) . Negative values indicate the proportion of movements discarded when moving from (N, M) to $(N, M+1)$ or $(N+1, M)$.

Fig. 2 summarizes these maps and reveals a marked asymmetry in hyperparameter influence. Increasing M is more restrictive: even for moderate values, +1 step typically discard approximately between -10% and -20% of remaining movements, with the sharpest reductions for $M \in [18, 28]$. In contrast, the impact of N remains limited with a constant effect for $N \leq 28$, where marginal losses remain below -13% . Sharp drops exceeding -50% occur only when N reaches extreme values ($N \geq 34$, (50% of total participants)).

These maps identify M as the dominant selectivity filter. Consequently, we fix N at the edge of its stability region ($N = 28$) and analyze how increasing M within its most selective range affects the retained movements per emotion.

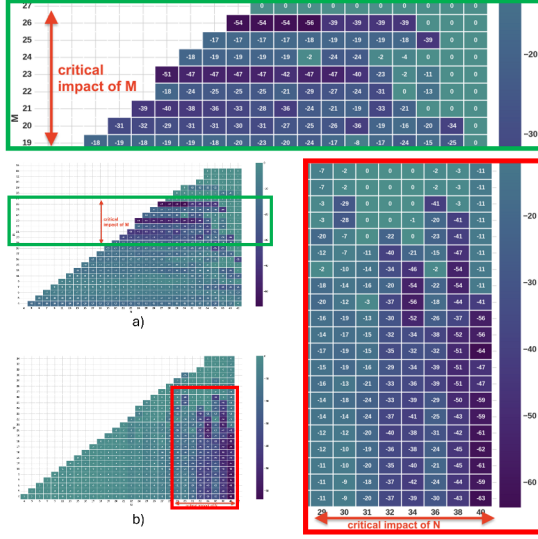


Figure 2: Local sensitivity maps of the retained movement volume. Panels display the percentage change (color gradient) due to a unit increase in (a) structural similarity threshold (Δ_M) and (b) temporal synchrony (Δ_N). Zoomed views highlight critical parameter ranges.

5.2 Taxonomy of Emotional Robustness

As shown in Fig. 3, raising M from 18 to 28 (from moderate to very strict structural similarity) causes a drastic reduction in movements, from 1039 to 44 and a loss of emotional diversity, dropping from ten emotions to only *Fear* for $M \geq 27$. Within this critical interval, emotions show clearly differentiated robustness profiles.

A first group – *Fear*, *Disgust* and *Anxiety* – forms a robust core. *Fear* is the most resilient, dominating the remaining movements at $M \geq 27$. *Disgust* and *Anxiety* show exceptional persistence, as they decay much more slowly than other emotions. Their motions are thus comparatively consistent across participants. A second group – *Amusement*, *Relief* and *Interest* – shows a high initial volume at $M = 18$ but declines and disappears around $M = 23$ -24. This implies that while these reactions generate high temporal synchrony motions, their execution manner is too variable to form a stable, highly homogeneous clusters under stricter thresholds. Finally, the low-robustness group (*Joy*, *Sadness*, *Tenderness* and *Anger*) is already sparse at $M = 18$ (except *Anger*), and disappears by $M = 21$, suggesting their bodily cues are either subtle or too idiosyncratic across participants to form a shared posture in passive viewing.

6 CONCLUSION AND FUTURE WORK

In this work, we addressed the scarcity of spontaneous 3D affective motion data by introducing a data-driven pipeline to isolate prototypical movements from raw recordings of standing participants in a passive viewing context. Instead of relying on manual selection, we proposed a consensus-based approach in which temporal synchrony (N) and structural similarity (M) act as complementary filters to remove parasitic motions and retain movements that are stimulus-induced and structurally shared across participants. Our preliminary analysis further shows that emotions do not withstand these constraints equally, highlighting an initial taxonomy that distinguishes stable “core” patterns from more idiosyncratic reactions.

Future work will extend this bottom-up framework toward explicit emotion differentiation. First, we will integrate finger-motion data (collected for 38 participants) to capture micro-gestural cues. Next, filtered movements will feed into a feature layer (e.g., kinematic, biomechanical, frequency-based descriptors) to assess inter-

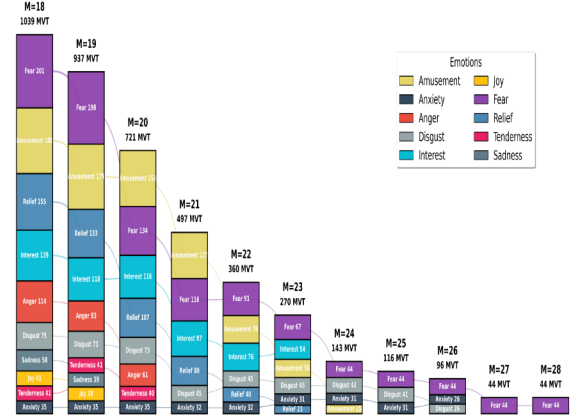


Figure 3: Evolution of Detected Movement Counts per Emotion as the parameter M Increases (Fixed $N = 28$, varying $M \in [18-28]$)

emotion separability via deep-learning clustering. Finally, a perceptual validation will be conducted with psychologists, progressively annotating prototypes from the highest (N, M) thresholds downwards to quantify how expressive validity varies with filter strength.

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REFERENCES

- [1] Advanced Realtime Tracking GmbH & Co. KG. *DTRACK3 User’s Guide*. Advanced Realtime Tracking GmbH & Co. KG, Weillheim i. OB, Germany, May 2023. 2
- [2] J. Aitchison. The statistical analysis of compositional data. *J. of the Royal Statistical Society: Series B (Methodological)*, 44(2), 1982. 3
- [3] H. Aviezer, Y. Trope, and A. Todorov. Body cues, not facial expressions, discriminate between intense positive and negative emotions. *Science*, 338:1225–1229, 2012. 1
- [4] T. Bänziger, H. Pirker, and K. R. Scherer. Gemep – geneva multimodal emotion portrayals: A corpus for the study of multimodal emotional expressions. In *Proceedings of the Fifth International Conference on Language Resources and Evaluation (LREC’06)*, 2000. 1
- [5] S. Buisine, M. Courgeon, A. Charles, C. Clavel, J.-C. Martin, N. Tan, and O. Grynszpan. The role of body postures in the recognition of emotions in contextually rich scenarios. *Int. J. of Human-Computer Interaction*, 30:52–62, 2014. 1
- [6] H. Chen, H. Shi, X. Liu, X. Li, and G. Zhao. Smg: A micro-gesture dataset towards spontaneous body gestures for emotional stress state analysis. *Int. J. Comput. Vision*, 131(6):1346–1366, Feb. 2023. 1
- [7] A. Crenn, A. Meyer, H. Konik, R. A. Khan, and S. Bouakaz. Generic body expression recognition based on synthesis of realistic neutral motion. *IEEE Access*, 8:207758–207767, 2020. 1
- [8] P. Ekman and W. Friesen. Detecting deception from the body or face. *J. of Personality and Social Psychology*, 29(3):288–298, 1974. 1
- [9] M. Greenacre. Compositional data analysis. *Annual Review of Statistics and Its Application*, 8:271–299, 2021. 3
- [10] R. Killick, P. Fearnhead, and I. A. Eckley. Optimal detection of changepoints with a linear computational cost. *J. of the American Statistical Association*, 107(500):1590–1598, Oct. 2012. 2
- [11] M.-A. Mahfoudi, A. Meyer, T. Gaudin, A. Buendia, and S. Bouakaz. Emotion expression in human body posture and movement: A survey on intelligible motion factors, quantification and validation. *IEEE Transactions on Affective Computing*, 14(4), 2023. 1
- [12] J. Russell. A circumplex model of affect. *J. of Personality and Social Psychology*, 39:1161–1178, 1980. 2